

The Impact of Limited-Edition Labeling on Consumer Purchasing Decisions

Group - 23

Rebecca Bubis, Sanjal Desai, Yashna Meher, Mishil Trivedi

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Introduction

The sneaker market has evolved significantly, transforming from functional footwear to a collector-driven industry fueled by exclusivity and speculative behavior. Our research explores whether labeling sneakers as "Limited Edition" impacts consumer purchase intent, willingness to pay (WTP), and resale interest. We hypothesized that consumers are more likely to purchase limited-edition sneakers, driven by perceived exclusivity and social signaling.

Research Question and Hypotheses

Research Question: Does labeling sneakers as "Limited Edition" influence consumers' purchasing behavior?

Null Hypothesis (H_0): Limited Edition labeling has no effect on consumer preferences, WTP, or purchase intent.

Alternative Hypothesis (H_1): Consumers exhibit higher purchase intent, WTP, and resale intent for Limited Edition labeled sneakers compared to regular ones.

Experimental Design

Our experiment involved a randomized online survey conducted via Qualtrics, recruiting participants primarily through social media. Participants were randomly assigned to either a control or a treatment group:

Control Group: Participants are explicitly informed which sneaker is a limited edition and which is a regular release when presented with images of the sneakers.

Treatment Group: Participants are shown images of both a limited edition and a regular release sneaker but are not provided with any information distinguishing between the two.

Each participant was randomly assigned to view either control or treatment conditions using Qualtrics' randomization features, ensuring unbiased results.

Survey Design

Participants in the control group were presented with images of two sneakers—one labeled as a regular release, the other as limited edition. Those in the treatment group received the same survey without the labels. The survey measured:

1. Purchase Intent - Consumers tend to show a stronger inclination toward limited edition releases due to perceived uniqueness and exclusivity. However, a portion remains neutral, focusing on personal style rather than availability.
2. Price Sensitivity - While premium materials justify a higher price for some (by 20% for this survey), many consumers resist paying more without exclusivity factors. Limited editions may slightly increase willingness to pay.
3. Past Shopping Behavior - Engagement in sneaker culture influences purchase behavior, with collectors more willing to pre-order or wait in line for exclusive releases. Casual buyers show less urgency.
4. Availability Influence - Limited stock can drive interest and urgency, making consumers more likely to choose the exclusive option. Always-available products may be perceived as less desirable.
5. Willingness to Pay - Consumers generally associate limited editions with higher value, making them more acceptable at premium price points, whereas regular releases face stricter price sensitivity.
6. Resale Intent - Limited edition sneakers attract speculative buyers who anticipate value appreciation, while regular releases see lower resale interest due to mass availability.
7. Choice Preference - Even at the same price, limited editions may maintain an edge in preference, reinforcing the psychological appeal of exclusivity in consumer decision-making.

Demographic and behavioral questions were included to identify heterogeneous effects based on gender, prior sneaker culture involvement, and general shopping behavior.

Data Analysis

Data was cleaned and processed for consistency. Missing values were and irrelevant data were discarded to maintain analytical clarity. We conducted several statistical analyses:

- Randomization Check (Proportions Z-Test): Confirmed that groups were randomized properly.
- Difference-in-Means Test (T-Test): Measured the average treatment effect (ATE).
- Regression Analysis: Explored deeper causal effects, controlling for demographic covariates.
- Cohen's D and Power Analysis: Calculated the effect size and assessed statistical power to interpret practical significance.

Results

Average Treatment Effect (ATE)

- Purchase Intent (1): The absence of labeling in the treatment group did not demonstrate a significant decrease in purchase intent compared to the control ($p > 0.05$).
- Resale Intent (2): A significant decrease in resale intent was found, suggesting exclusivity with “Limited Edition” encourages speculative purchasing behaviors ($p < 0.01$).
- WTP (3): Participants in the treatment group displayed a slightly positive increase in paying more for shoes without knowing which is limited edition, indicating price insensitivity to exclusivity; however, this was not found to be statistically significant ($p > 0.05$).

	(1)	(2)	(3)
Intercept	1.500*** (0.067)	1.679*** (0.065)	1.411*** (0.067)
treatment	-0.027 (0.096)	-0.279*** (0.092)	0.044 (0.095)
Observations	111	111	111
R ²	0.001	0.078	0.002
Adjusted R ²	-0.008	0.070	-0.007
Residual Std. Error	0.504 (df=109)	0.483 (df=109)	0.499 (df=109)
F Statistic	0.081 (df=1; 109)	9.235*** (df=1; 109)	0.214 (df=1; 109)
Note:	*p<0.1; **p<0.05; ***p<0.01		

Heterogeneous Treatment Effects

We found demographic variations in treatment effects:

1. Gender (1): Males exhibited significantly higher purchase intent for limited edition sneakers compared to females, which could be explained by the product type used for this survey with men historically being more interested in sneakers than women ($p < 0.05$).(iii)

2. Past Shopping Behavior (2): Participants previously involved in sneaker culture demonstrated weaker, speculative interest, reflecting no significant change in their urgency to purchase when not exposed to Limited Edition cues ($p > 0.05$).

	(1)	(2)
C(gender)[T.Male]	0.233** (0.094)	
Intercept	1.367*** (0.085)	1.357*** (0.063)
treatment	-0.025 (0.094)	-0.048 (0.090)
Observations	111	111
R ²	0.054	0.003
Adjusted R ²	0.037	-0.007
Residual Std. Error	0.493 (df=108)	0.475 (df=109)
F Statistic	3.085** (df=2; 108)	0.284 (df=1; 109)
Note:	* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$	

Interpretation of Results

Our findings reveal the nuanced role of exclusivity. Contrary to initial expectations, limited-edition labels alone were insufficient to boost purchase intent significantly. However, they did significantly enhance speculative behaviors (resale intent)(iii). This aligns with other research demonstrating how speculative and emotional motivators (e.g., fear of missing out - FOMO) strongly influence purchasing decisions, even without affecting direct purchase intent.(i)

The insignificant direct impact on purchase intent contrasts with prior assumptions about exclusivity as a primary driver of sneaker sales. The observed speculative behavior confirms exclusivity's indirect value proposition, primarily through resale market appeal.

Limitations and Future Research

This experiment had several limitations:

1. Sample Bias: Demographic skewness towards university students potentially limits generalizability.
2. Small Sample Size: Our 55 observations per group (after data cleaning) limited the experiment's statistical power (Cohen's $d = -0.054$, power ≈ 0.059), making it challenging to detect subtle treatment effects robustly.(ii)
3. Duration of Experiment: A short data collection window (one week) limited the ability to capture broader temporal trends, potentially biasing outcomes by excluding typical shopping variability.

Future research should address these limitations by employing larger, diverse samples, extended observation periods, and exploring deeper psychological factors that drive speculative buying behaviors.

Business Implications and Recommendations

Our experiment provides practical insights:

- Marketing Implication: Limited Edition labels primarily drive speculative behavior, suggesting strategic utility in targeting collector communities rather than general consumers.
- Segmented Targeting: Marketers should strategically use exclusivity cues to engage demographics responsive to speculative value—primarily sneaker collectors and males.
- Comprehensive Marketing Strategies: Businesses should adopt holistic strategies, incorporating exclusivity within broader promotional frameworks rather than relying solely on scarcity labels to drive direct consumer purchases.

Conclusion

The experiment aimed to dissect the real-world impact of labeling sneakers as "Limited Edition" on consumer purchasing decisions. While exclusivity labels did not significantly alter direct purchase intent or willingness to pay, they enhanced speculative behavior, highlighting exclusivity's indirect appeal through resale value(iii). These results underscore the complexity of consumer decision-making processes and the essential role of strategic demographic targeting in exclusivity-based marketing campaigns. Although social media and word-of-mouth marketing have contributed to the widespread belief that labeling a product as "Limited Edition" naturally drives higher sales, our experiment has yielded results that challenge this assumption. Our findings indicate that simply adding a "Limited Edition" label does not inherently lead to an increase in sales. Instead, its effectiveness is highly dependent on a well-executed targeting strategy. To maximize the impact of such labeling, businesses must carefully identify and engage the right audience, leveraging exclusivity as a compelling value proposition rather than assuming it will automatically enhance demand.

Future research should consider extended timelines, larger and more diverse samples, and deeper explorations into consumer psychology, contributing to more robust findings and actionable marketing insights. Ultimately, the strategic use of exclusivity remains vital but requires precise implementation tailored to specific consumer segments and clearly defined marketing objectives.

This research deepens the understanding of exclusivity cues in marketing and provides clear strategic directions for brands operating in high-competition consumer markets such as sneakers. For more information about this project, please refer to [this folder](#).

References

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